

ticker	parent_company	start_year	end_year	product_family	product_name	search_term
TMO	Thermo Fisher Scientific			Orbitrap	Orbitrap Exploris/Ascend/Eclipse/Fusion/Astral/Elite	Orbitrap
TMO	Thermo Fisher Scientific			Q Exactive	Q Exactive HF/Plus/UHMR/Focus	Q Exactive
TMO	Thermo Fisher Scientific			Exactive	Exactive Plus/EMR	Exactive
TMO	Thermo Fisher Scientific			Astral	Orbitrap Astral	Astral
TMO	Thermo Fisher Scientific			TSQ	TSQ Altis/Quantis/Fortis/Endura/Plus	TSQ
TMO	Thermo Fisher Scientific			ISQ	ISQ EC/EM/7000 GC-MS	ISQ
TMO	Thermo Fisher Scientific			Stellar	Stellar MS	Stellar MS
TMO	Thermo Fisher Scientific			Neoma	Neoma multicollector ICP-MS	Neoma
TMO	Thermo Fisher Scientific			Element MS	Element XR/2 HR-ICP-MS	Element XR
TMO	Thermo Fisher Scientific			TriPlus	TriPlus RSH autosampler	TriPlus
TMO	Thermo Fisher Scientific			Direct Mass Technology	Direct Mass Technology (charge detection)	Direct Mass Technology
TMO	Thermo Fisher Scientific			Vanquish	Vanquish Horizon/Flex/Neo/Core UHPLC	Vanquish
TMO	Thermo Fisher Scientific			UltiMate	UltiMate 3000 HPLC	UltiMate 3000
TMO	Thermo Fisher Scientific			Dionex ICS	Dionex ICS ion chromatography	Dionex ICS
TMO	Thermo Fisher Scientific			TRACE	TRACE 1300/1310 GC	TRACE 1300
TMO	Thermo Fisher Scientific			Acclaim	Acclaim HPLC columns	Acclaim
TMO	Thermo Fisher Scientific			Hypersil	Hypersil GOLD HPLC columns	Hypersil
TMO	Thermo Fisher Scientific			EASY-Spray	EASY-Spray / EASY-nLC nano-LC	EASY-nLC
TMO	Thermo Fisher Scientific			Nicolet	Nicolet iS20/iS50/Summit FTIR	Nicolet
TMO	Thermo Fisher Scientific			iCAP	iCAP RQ/TQ/PRO ICP-MS/OES	iCAP
TMO	Thermo Fisher Scientific			iCE	iCE 3000 AAS	iCE 3000
TMO	Thermo Fisher Scientific			NanoDrop	NanoDrop spectrophotometer	NanoDrop
TMO	Thermo Fisher Scientific			Multiskan	Multiskan microplate photometer	Multiskan
TMO	Thermo Fisher Scientific			Varioskan	Varioskan LUX multimode reader	Varioskan
TMO	Thermo Fisher Scientific			Fluoroskan	Fluoroskan microplate fluorometer	Fluoroskan
TMO	Thermo Fisher Scientific			Luminoskan	Luminoskan luminometer	Luminoskan

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PubMed 2025: 36%[REDACTED] | 14/39 [01:07<01:05, 2.60s/it, failures=0, processed=4095]2026-07-24 14:13:58,752 INFO PubMed 2025 terms_015 segment 1: 2025-01-01 to 2025-12-31, 833 candidate PMIDs
PubMed 2025: 38%[REDACTED] | 15/39 [01:15<01:40, 4.20s/it, failures=0, processed=4843]2026-07-24 14:14:06,430 INFO PubMed 2025 terms_016 segment 1: 2025-01-01 to 2025-12-31, 111 candidate PMIDs
PubMed 2025: 41%[REDACTED] | 16/39 [01:16<01:17, 3.38s/it, failures=0, processed=4942]2026-07-24 14:14:07,914 INFO PubMed 2025 terms_017 segment 1: 2025-01-01 to 2025-12-31, 45 candidate PMIDs
PubMed 2025: 44%[REDACTED] | 17/39 [01:17<00:56, 2.58s/it, failures=0, processed=4978]2026-07-24 14:14:08,786 INFO PubMed 2025 terms_018 segment 1: 2025-01-01 to 2025-12-31, 452 candidate PMIDs
PubMed 2025: 46%[REDACTED] | 18/39 [01:21<01:02, 3.00s/it, failures=0, processed=5484]2026-07-24 14:14:12,584 INFO PubMed 2025 terms_019 segment 1: 2025-01-01 to 2025-12-31, 79 candidate PMIDs
PubMed 2025: 49%[REDACTED] | 19/39 [01:22<00:48, 2.43s/it, failures=0, processed=5477]2026-07-24 14:14:13,725 INFO PubMed 2025 terms_020 segment 1: 2025-01-01 to 2025-12-31, 273 candidate PMIDs
PubMed 2025: 51%[REDACTED] | 20/39 [01:25<00:47, 2.52s/it, failures=0, processed=5730]2026-07-24 14:14:17,193 INFO PubMed 2025 terms_021 segment 1: 2025-01-01 to 2025-12-31, 2442 candidate PMIDs
PubMed 2025: 54%[REDACTED] | 21/39 [01:48<02:35, 8.62s/it, failures=0, processed=8082]2026-07-24 14:14:39,772 INFO PubMed 2025 terms_022 segment 1: 2025-01-01 to 2025-12-31, 484 candidate PMIDs
PubMed 2025: 56%[REDACTED] | 22/39 [01:52<02:05, 7.37s/it, failures=0, processed=8161]2026-07-24 14:14:45,339 WARNING pubmed HTTP 502 (attempt 1/7); waiting 3.4s
2026-07-24 14:14:50,222 INFO PubMed 2025 terms_023 segment 1: 2025-01-01 to 2025-12-31, 2907 candidate PMIDs
PubMed 2025: 59%[REDACTED] | 23/39 [02:28<04:12, 15.77s/it, failures=0, processed=10445]2026-07-24 14:15:19,336 INFO PubMed 2025 terms_024 segment 1: 2025-01-01 to 2025-12-31, 1339 candidate PMIDs
PubMed 2025: 62%[REDACTED] | 24/39 [02:39<03:36, 14.42s/it, failures=0, processed=11347]2026-07-24 14:15:30,400 INFO PubMed 2025 terms_025 segment 1: 2025-01-01 to 2025-12-31, 404 candidate PMIDs
PubMed 2025: 64%[REDACTED] | 25/39 [02:43<02:39, 11.38s/it, failures=0, processed=11725]2026-07-24 14:15:34,725 INFO PubMed 2025 terms_026 segment 1: 2025-01-01 to 2025-12-31, 440 candidate PMIDs
PubMed 2025: 67%[REDACTED] | 26/39 [02:47<01:59, 9.19s/it, failures=1, processed=12004]2026-07-24 14:15:39,008 INFO PubMed 2025 terms_027 segment 1: 2025-01-01 to 2025-12-31, 226 candidate PMIDs
PubMed 2025: 69%[REDACTED] | 27/39 [02:50<01:26, 7.25s/it, failures=3, processed=12152]2026-07-24 14:15:41,661 INFO PubMed 2025 terms_028 segment 1: 2025-01-01 to 2025-12-31, 78 candidate PMIDs
PubMed 2025: 72%[REDACTED] | 28/39 [02:52<01:01, 5.55s/it, failures=3, processed=12200]2026-07-24 14:15:43,796 INFO PubMed 2025 terms_029 segment 1: 2025-01-01 to 2025-12-31, 195 candidate PMIDs
PubMed 2025: 74%[REDACTED] | 29/39 [02:54<00:47, 4.70s/it, failures=3, processed=12341]2026-07-24 14:15:46,102 INFO PubMed 2025 terms_030 segment 1: 2025-01-01 to 2025-12-31, 478 candidate PMIDs
PubMed 2025: 77%[REDACTED] | 30/39 [03:01<00:47, 5.24s/it, failures=3, processed=12430]2026-07-24 14:15:52,892 INFO PubMed 2025 terms_031 segment 1: 2025-01-01 to 2025-12-31, 132 candidate PMIDs
PubMed 2025: 79%[REDACTED] | 31/39 [03:05<00:38, 4.83s/it, failures=4, processed=12516]2026-07-24 14:15:56,972 INFO PubMed 2025 terms_032 segment 1: 2025-01-01 to 2025-12-31, 199 candidate PMIDs
PubMed 2025: 82%[REDACTED] | 32/39 [03:07<00:29, 4.20s/it, failures=5, processed=12674]2026-07-24 14:16:00,297 INFO PubMed 2025 terms_033 segment 1: 2025-01-01 to 2025-12-31, 404 candidate PMIDs
PubMed 2025: 85%[REDACTED] | 33/39 [03:12<00:26, 4.41s/it, failures=5, processed=12906]2026-07-24 14:16:04,702 INFO PubMed 2025 terms_034 segment 1: 2025-01-01 to 2025-12-31, 174 candidate PMIDs
PubMed 2025: 87%[REDACTED] | 34/39 [03:15<00:19, 3.92s/it, failures=5, processed=12994]2026-07-24 14:16:07,594 INFO PubMed 2025 terms_035 segment 1: 2025-01-01 to 2025-12-31, 1521 candidate PMIDs
PubMed 2025: 90%[REDACTED] | 35/39 [03:32<00:30, 7.70s/it, failures=5, processed=13771]2026-07-24 14:16:23,850 INFO PubMed 2025 terms_036 segment 1: 2025-01-01 to 2025-12-31, 1192 candidate PMIDs
PubMed 2025: 92%[REDACTED] | 36/39 [03:47<00:30, 10.06s/it, failures=5, processed=14799]2026-07-24 14:16:38,910 INFO PubMed 2025 terms_037 segment 1: 2025-01-01 to 2025-12-31, 1453 candidate PMIDs
PubMed 2025: 95%[REDACTED] | 37/39 [04:03<00:23, 11.83s/it, failures=5, processed=15339]2026-07-24 14:16:54,777 INFO PubMed 2025 terms_038 segment 1: 2025-01-01 to 2025-12-31, 1169 candidate PMIDs
PubMed 2025: 97%[REDACTED] | 38/39 [04:14<00:11, 11.48s/it, failures=5, processed=16404]2026-07-24 14:17:05,426 INFO PubMed 2025 terms_039 segment 1: 2025-01-01 to 2025-12-31, 191 candidate PMIDs
PubMed 2025: 100%[REDACTED] | 39/39 [04:18<00:00, 6.62s/it, failures=5, processed=16563]
2026-07-24 14:17:13,352 INFO Exported pubmed 2025: 15267 company-record rows across 3 match file(s)

```

851 enabled terms across 23 companies; matcher: ahocorasick

11 enabled term(s) are <=3 chars and case-insensitive (high false-positive risk): ['BEH', 'CFX', 'DxH', 'EZ1', 'ISQ', 'QDa', 'S3e', 'TGX', 'TSQ', 'VWR', 'ZE5']

RUN LABEL

2022

FROM DATE

01/01/2022

TO DATE

12/31/2022

RETRIEVAL MODE

JATS XML only — recommended

SECTIONS INCLUDED IN RESULTS

Core article only — recommended

SAMPLING MODE

Random order until N valid full texts

RANDOM SEED

2022

VALID FULL-TEXT TARGET

5000

MAXIMUM PAPERS ATTEMPTED

7500

A paper counts only when its JATS XML (or HTML fallback, if selected) is successfully retrieved and parsed. Failed papers stay in the audit log and are replaced by the next paper in the pre-saved random order. The main results CSV contains successful full-text papers only.

JATS DELAY MINIMUM (SECONDS)

10.0

JATS DELAY MAXIMUM (SECONDS)

20.0

API DELAY MINIMUM

1.0

API DELAY MAXIMUM

2.0

☐ Hide peripheral matches from the live feed when using a broader section scope

Start or resume run

Use the same run label, dates, seed, target, maximum attempts, dictionaries, and retrieval settings to resume. Change the run label to intentionally create a separate fresh

```

def collect_random_sample(client, start, end, sample_size, sample_seed,
                          on_progress=None):
    all_records = []
    for _, _, rec in client.enumerate_window(
        start, end, cursor=0, on_progress=on_progress, max_papers=None):
        all_records.append(rec)
    population = _dedupe_latest(all_records)
    if sample_size is None or sample_size >= len(population):
        selected = list(population)
    else:
        selected = random.Random(sample_seed).sample(population, sample_size)
    selected.sort(key=lambda rec: (rec.get("doi", "").lower(),
                                     _version_number(rec.get("version"))))
    if on_progress:
        on_progress({
            "phase": "sample",
            "population": len(population),
            "selected": len(selected),
            "seed": sample_seed,
        })
    return selected, len(population)

```

```

def collect_random_order(client, start, end, sample_seed,
                          on_progress=None):
    """Return every unique DOI in one reproducible random order.

    This is used by target-success mode. The full order is saved before any
    full-text retrieval begins, so failed papers are replaced only by the next
    paper in a precommitted random ordering rather than by an ad-hoc redraw.
    """
    all_records = []
    for _, _, rec in client.enumerate_window(
        start, end, cursor=0, on_progress=on_progress, max_papers=None):
        all_records.append(rec)
    population = _dedupe_latest(all_records)
    random.Random(sample_seed).shuffle(population)
    ordered = []
    for rank, rec in enumerate(population, start=1):
        item = dict(rec)
        item["random_rank"] = rank
        ordered.append(item)
    if on_progress:
        on_progress({
            "phase": "sample",
            "population": len(ordered),
            "selected": len(ordered),
            "seed": sample_seed,
            "ordered_population": True,
        })
    return ordered, len(ordered)

```